Descriptor Similarity Does Not Predict Array Response: Chemical Warfare Agent Simulants Tested Against Seventeen Authentic Agents

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Abstract

Chemical warfare agent detection is developed and qualified on simulants, and we asked what measurement establishes that a simulant reproduces the agent it stands in for. A 29-dye colorimetric array was used to measure seventeen authentic agents, spanning nerve, blood, choking, vesicant and A-series compounds, at five concentrations in two optical channels, and the resulting response patterns were compared with the molecular-descriptor distances on which simulant selection in practice rests. Agreement was tested by exact Mantel permutation within each run, by exact enumeration of all 680 possible simulant designations in descriptor space, and against permutation nulls built for each comparison. The array separates compounds in all six run-channel cells, with p from 4.6×10^{-8} to 0.025, so the measurement is sound. Descriptor distance nonetheless fails to predict response distance: twelve exact Mantel tests and six

combinations of them leave nothing below $q = 0.05$. Dimethyl methylphosphonate, the standard G-agent simulant and the benchmark compound for decontamination efficiency, is anti-correlated with soman and cyclosarin in the ultraviolet channel, and its correlation with sarin falls from $+0.77$ at 500 ppm to $+0.36$ at 10 ppm, the end at which point detectors operate. Across the seventeen agents measured, exactly one designated simulant has ever been measured beside them.

Introduction

Nearly every chemical warfare agent detector in the literature is developed, optimised and qualified on compounds that are not chemical warfare agents.[1,2] Dimethyl methylphosphonate stands in for the G-series nerve agents,[3] dipropylene glycol monomethyl ether for training detectors,[4] and methyl salicylate for vapour penetration through protective ensembles.[5,6] The substitution is unavoidable, because the agents are lethal at concentrations where handling them requires a licensed facility, and it is load-bearing: a sensor reported as selective for sarin has, in most cases, never been exposed to sarin.

What licenses the substitution is a claim of similarity. The compound chosen is said to be agent-like, and similarity is read from molecular descriptors: vapour pressure, dipole moment, frontier orbital energies, substructure and shape. The claim that matters for a detector is different. It is that the substitute produces the response the agent would produce in the same device. Those two claims are connected only if descriptor similarity transfers to response similarity, and that transfer has not been measured.

It has not been measured because measuring it requires the agents. Response data on authentic agents exist in few places, and the compounds used as simulants are rarely run alongside them. A recent survey of the field found no standard for chemical warfare agent simulants at all, and recorded that they have generally been selected on the basis of their historical use.[7] Our institute is an OPCW designated laboratory, which identifies and assays authentic agents under an accredited procedure, so the comparison can be made on the

agents themselves.

This work reports that measurement. A 29-dye colorimetric array was used to record the responses of seventeen authentic agents at five concentrations in two optical channels, and those responses were compared against the molecular-descriptor distances that simulant selection relies on. The array separates compounds in every run and channel, so the measurement is sound. Descriptor distance nonetheless carries no detectable information about response distance. Dimethyl methylphosphonate, the compound the field treats as the G-agent surrogate, is anti-correlated with two of the four G agents, and its resemblance to sarin weakens as the concentration falls towards the range where point detectors operate. Where the designation record and a regulatory toxicity model might have supplied the missing evidence, both are empty for the same compounds.

This is not a new sensing method, and we propose no replacement compound. It is a measurement of whether the substitutes the field already relies on carry the property they are relied upon for.

Results and Discussion

The array separates the agents

Descriptor distance does not predict response distance

The standard G-agent simulant against the G agents

Descriptor space cannot supply the missing validation

Neither the designation record nor a toxicity model fills the gap

Conclusions

Experimental Section

Acknowledgement

Supporting Information Available

References

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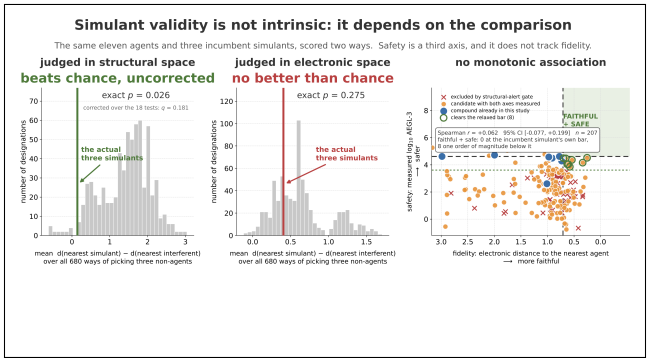
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TOC Graphic


Simulant validity is not intrinsic: it depends on the comparison
The same eleven agents and three incumbent simulants, scored two ways. Safety is a third axis, and it does not track fidelity.
judged in structural space
beats chance, uncorrected
exact p = 0.026
corrected over the 18 tests: q = 0.181
the actual three simulants
number of designations
mean d(nearest simulant) − d(nearest interferent) over all 680 ways of picking three non-agents
judged in electronic space
no better than chance
exact p = 0.275
the actual three simulants
number of designations
mean d(nearest simulant) − d(nearest interferent) over all 680 ways of picking three non-agents
no monotonic association
excluded by structural-alert gate
candidate with both axes measured
compound already in this study
clears the relaxed bar (8)
FAITHFUL + SAFE
Spearman r = +0.062 95% CI [-0.077, +0.198] n = 207
faithful + safe: 0 at the incumbent simulant's own bar;
8 nine order of magnitude below it
safety: measured log10 AEGL-3
fidelity: electronic distance to the nearest agent
→ more faithful